3.1 Introduction



Let us consider the given situation that Ganesh planted saplings in his garden. He planted 10 rows each with 5 saplings. Can you say how many saplings were planted?

Yes, we know that, the total number of saplings is the product of number of rows and number of saplings in each row.

Hence, the total number of saplings = 10 rows $\times$ 5 saplings in each row $= 10 \times 5 = 50$ saplings

Likewise, David planted some saplings. Not knowing the total number of rows and saplings in each row, how will you express the total number of saplings?

For the unknown quantities, we call them as 'x' and 'y'. Therefore, the total number of saplings = 'x' rows $\times$ 'y' sapling in each row

= '$x \times y$' = xy saplings

Let us extend this situation, Rahim planted saplings where the number of rows are $(2x^2 + 5x - 7)$ and each row contains $3y^2$ saplings. Now the above idea will help us to find the total number of saplings planted by Rahim.

The total number of saplings = $(2x^2 + 5x - 7)$ rows $\times$ $3y^2$ saplings in each row.

$$= 3y^2 \times (2x^2 + 5x - 7)$$

How do we find the product of the above algebraic expression?

Now,we will learn to find the product of algebraic expressions.

Note

Polynomial	A polynomial is an expression containing two or more algebraic terms. In a polynomial all variables are raised to only whole number powers. $a^2 + 2ab + b^2$ $\quad$ $4x^2 + 3x - 7$ A polynomial cannot contain : 1) Division by a variable. Eg. $4x^2 - \frac{5}{1+x}$ is not a polynomial. 2) Negative exponents. Eg. $7x^{-2} + 5x - 6$ is not a polynomial. 3) Fractional exponents. Eg. $3x^3 + 4x^{\frac{1}{2}} + 5$ is not a polynomial.
Monomial	An expression which contains only one term is called a monomial. Examples: $4x$, $3x^2y$, $-2y^2$.
Binomial	An expression which contains only two terms is called a binomial. Examples: $2x+3$, $5y^2+9y$, a^2b^2+2b .
Trinomial	An expression which contains only three terms is called a trinomial. Examples : $2a^2b-8ab+b^2$, m^2-n^2+3 .

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